

CAREBRIDGE HEALTH NETWORK

Hospital Performance, Financial and Operational Analytics Report

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Microsoft Excel//MySQL// Power BI

Analysis Period: January 2023 – December 2023.

1.0 Introduction

The healthcare industry generates large volumes of data daily through patient interactions, clinical activities, appointments, treatments, and financial transactions. However, the true value of this data lies not in its collection but in its ability to support informed decision-making.

CareBridge Health Network operates multiple healthcare facilities that provide a wide range of medical services to patients across different demographic groups. As healthcare demand continues to increase, management requires accurate and timely insights to effectively allocate resources, improve operational efficiency, and maintain financial sustainability.

This project was undertaken to transform raw operational data into meaningful business intelligence capable of supporting strategic and operational decisions. Through the application of data analytics techniques, key performance indicators were measured and analyzed to evaluate organizational performance across patient management, doctor performance, treatment utilization, appointment management, and financial operations.

The analysis was performed using SQL for data preparation and Power BI for dashboard development and visualization. By integrating data from multiple operational sources, the project provides a comprehensive view of hospital performance and highlights areas where improvements can drive measurable business impact.

2.0 Executive Summary

This report presents an analysis of the operational and financial performance of **CareBridge Health Network**, a healthcare organization comprising **Central Hospital**, **Eastside Clinic**, and **Westside Clinic**. The analysis covers the period from **January to December 2023** and was conducted using **Microsoft Excel** for data cleaning, **MySQL** for data storage and querying, and **Microsoft Power BI** for visualization and dashboard development.

During the period under review, CareBridge Health Network generated a **total revenue of \$551,250** from **200 appointments**, **200 treatments**, and **50 registered patients**, supported by a team of **10 medical doctors** across its three facilities.

While the organization demonstrated a strong revenue base, the analysis identified several operational challenges affecting overall performance. Only **23% of appointments were completed**, indicating that a significant proportion of scheduled appointments ended as cancellations, no-shows, or remained pending. Furthermore, the hospital achieved a **payment collection rate of only 31.46%**, suggesting that a substantial portion of billed revenue remained unpaid.

The analysis also revealed that **Central Hospital** was the highest-performing facility, contributing approximately **\$229,000 (41.6%)** of total revenue. In terms of service lines, **Chemotherapy (\$129,000)** generated the highest revenue, followed by **MRI (\$116,000)** and **X-Ray (\$111,000)**. These findings indicate that oncology and diagnostic imaging services are the primary financial drivers of the network.

Overall, the findings suggest that CareBridge Health Network possesses strong revenue-generating potential but is constrained by operational inefficiencies rather than structural limitations. Improvements in appointment management, payment collection, patient engagement, and resource allocation present significant opportunities to increase revenue without expanding infrastructure or workforce.

3.0 Project Objectives

The primary objective of this project is to evaluate the overall performance of CareBridge Health Network and identify opportunities for operational and financial improvement.

Specific objectives include:

1. To assess the hospital network's overall operational performance.
2. To analyze patient demographics and registration patterns.
3. To evaluate doctor performance and specialization trends.
4. To examine appointment scheduling and completion rates.
5. To identify the treatments contributing most significantly to revenue generation.
6. To evaluate billing performance and payment collection efficiency.
7. To uncover operational bottlenecks affecting organizational performance.
8. To provide data-driven recommendations for improving service delivery, patient experience, and financial outcomes.

4.0 Scope and Timeframe of Analysis

This analysis examines operational and financial activities across CareBridge Health Network over a twelve-month period spanning January [YEAR] to December [YEAR].

The project incorporates data from five core business functions:

1. Patient Management
2. Doctor Management
3. Appointment Management
4. Treatment Management
5. Billing and Revenue Management

The dataset analyzed consists of:

1. 50 registered patients
2. 10 doctors
3. 200 appointments
4. 200 treatment records
5. Associated billing transactions across all facilities

The analysis focuses on identifying trends, performance patterns, operational challenges, and revenue opportunities across the organization during the reporting period.

5.0 Data Sources and Structure

The analysis was conducted using five interconnected datasets representing key operational functions within CareBridge Health Network.

Patients Table

Contains patient demographic and registration information, including gender, age, contact details, registration dates, and insurance information.

	patient_id	first_name	last_name	gender	date_of_birth	contact_number	address	registration_date	insurance_provider	insurance_number	email
▶	P001	David	Williams	F	2022-06-23	6939585183	789 Pine Rd	2022-06-23	WellnessCorp	INS840674	david.williams@mail.com
	P002	Emily	Smith	F	2022-01-15	8228188767	321 Maple Dr	2022-01-15	PulseSecure	INS354079	emily.smith@mail.com
	P003	Laura	Jones	M	2022-02-07	8397029847	321 Maple Dr	2022-02-07	PulseSecure	INS650929	laura.jones@mail.com
	P004	Michael	Johnson	F	2021-03-02	9019443432	123 Elm St	2021-03-02	HealthIndia	INS789944	michael.johnson@mail.com
	P005	David	Wilson	M	2021-09-29	7734463155	123 Elm St	2021-09-29	MedCare Plus	INS788105	david.wilson@mail.com

Doctors Table

Contains doctor information such as specialization, years of experience, branch assignment, and professional contact details.

	doctor_id	first_name	last_name	specialization	phone_number	years_experience	hospital_branch	email
▶	D001	David	Taylor	Dermatology	8322010158	17	Westside Clinic	dr.david.taylor@hospital.com
	D002	Jane	Davis	Pediatrics	9004382050	24	Eastside Clinic	dr.jane.davis@hospital.com
	D003	Jane	Smith	Pediatrics	8737740598	19	Eastside Clinic	dr.jane.smith@hospital.com
	D004	David	Jones	Pediatrics	6594221991	28	Central Hospital	dr.david.jones@hospital.com
	D005	Sarah	Taylor	Dermatology	9118538547	26	Central Hospital	dr.sarah.taylor@hospital.com
	D006	Alex	Davis	Pediatrics	6570137231	23	Central Hospital	dr.alex.davis@hospital.com

Appointments Table

Contains appointment records including appointment dates, assigned doctors, patient information, reasons for visit, and appointment status.

	appointment_id	patient_id	doctor_id	appointment_date	appointment_time	reason_for_visit	status
▶	A001	P034	D009	2023-08-09	3:15:00 PM	Therapy	Scheduled
	A002	P032	D004	2023-06-09	2:30:00 PM	Therapy	No-show
	A003	P048	D004	2023-06-28	8:00:00 AM	Consultation	Cancelled
	A004	P025	D006	2023-09-01	9:15:00 AM	Consultation	Cancelled
	A005	P040	D003	2023-07-06	12:45:00 PM	Emergency	No-show
	A006	P045	D006	2023-06-19	4:15:00 PM	Checkup	Scheduled
	A007	P001	D007	2023-04-09	10:30:00 AM	Consultation	Scheduled
	A008	P016	D010	2023-05-24	8:45:00 AM	Consultation	Cancelled
	A009	P039	D010	2023-03-05	1:45:00 PM	Follow-up	Scheduled
	A010	P005	D003	2023-01-13	3:30:00 PM	Therapy	Completed

Treatments Table

Contains treatment information including treatment type, treatment date, treatment description, and treatment costs.

	treatment_id	appointment_id	treatment_type	description	cost	treatment_date
▶	T001	A001	Chemotherapy	Basic screening	3941.97	2023-08-09
	T002	A002	MRI	Advanced protocol	4158.44	2023-06-09
	T003	A003	MRI	Standard procedure	3731.55	2023-06-28
	T004	A004	MRI	Basic screening	4799.86	2023-09-01
	T005	A005	ECG	Standard procedure	582.05	2023-07-06
	T006	A006	Chemotherapy	Standard procedure	1381.00	2023-06-19
	T007	A007	Chemotherapy	Advanced protocol	534.03	2023-04-09
	T008	A008	Physiotherapy	Basic screening	3413.64	2023-05-24
	T009	A009	Physiotherapy	Standard procedure	4541.14	2023-03-05
	T010	A010	Physiotherapy	Standard procedure	1595.67	2023-01-13

Billing Table

Contains financial transaction records including billing dates, payment methods, payment status, and revenue generated from treatments.

	bill_id	patient_id	treatment_id	bill_date	amount	payment_method	payment_status
▶	B001	P034	T001	2023-08-09	3941.97	Insurance	Pending
	B002	P032	T002	2023-06-09	4158.44	Insurance	Paid
	B003	P048	T003	2023-06-28	3731.55	Insurance	Paid
	B004	P025	T004	2023-09-01	4799.86	Insurance	Failed
	B005	P040	T005	2023-07-06	582.05	Credit Card	Pending
	B006	P045	T006	2023-06-19	1381.00	Insurance	Pending
	B007	P001	T007	2023-04-09	534.03	Cash	Failed
	B008	P016	T008	2023-05-24	3413.64	Cash	Failed
	B009	P039	T009	2023-03-05	4541.14	Credit Card	Paid
	B010	P005	T010	2023-01-13	1595.67	Cash	Paid

The relationships among these tables enabled comprehensive analysis of patient activity, doctor performance, treatment utilization, appointment outcomes, and revenue generation across the network.

6.0 Data Cleaning and Preparation

Prior to analysis, the dataset underwent a structured data cleaning and preparation process to ensure data quality, consistency, and reliability.

The following activities were performed:

Data Validation

Each table was reviewed to identify inconsistencies, duplicate records, missing values, and formatting issues.

Data Type Standardization

Date fields including:

1. Date of Birth
2. Registration Date
3. Appointment Date
4. Treatment Date
5. Billing Date

were converted into appropriate date formats to support time-based analysis.

Numeric fields such as treatment costs and billing amounts were standardized to enable accurate financial calculations.

Missing Value Handling

Columns containing entirely null values were removed where appropriate, while records with critical missing information were reviewed and addressed to maintain data integrity.

Relationship Validation

Primary and foreign key relationships between tables were validated to ensure accurate data modeling and reporting within Power BI.

Data Modeling

The cleaned data was imported into Power BI and structured using a relational model linking patients, doctors, appointments, treatments, and billing records.

This model enabled cross-functional analysis and supported the development of interactive dashboards used throughout this report.

7.0 Executive Overview Analysis

7.1 Overall Organizational Performance

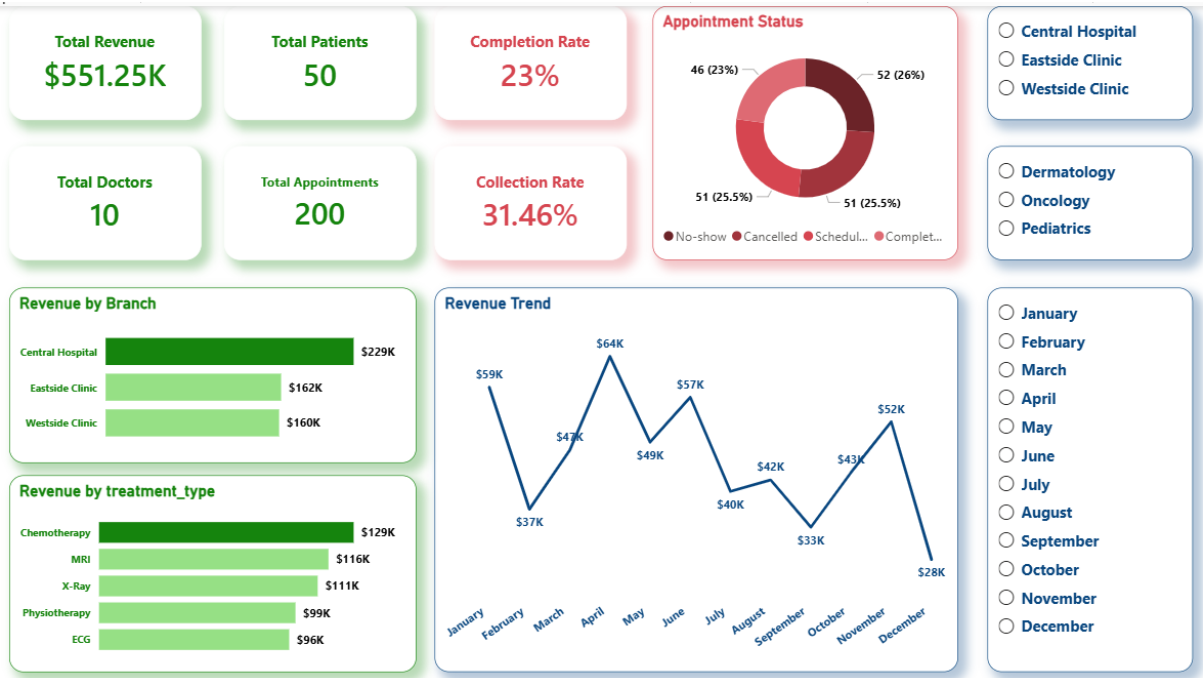
The Executive Overview dashboard provides a high-level assessment of CareBridge Health Network's operational and financial performance across the reporting period.

During the analysis period, the network generated approximately **\$551.25K in revenue**, served **50 patients**, managed **200 appointments**, and employed **10 doctors** across its three healthcare facilities.

While revenue performance remained relatively strong throughout the year, key operational indicators reveal opportunities for improvement, particularly in appointment management and revenue collection processes.

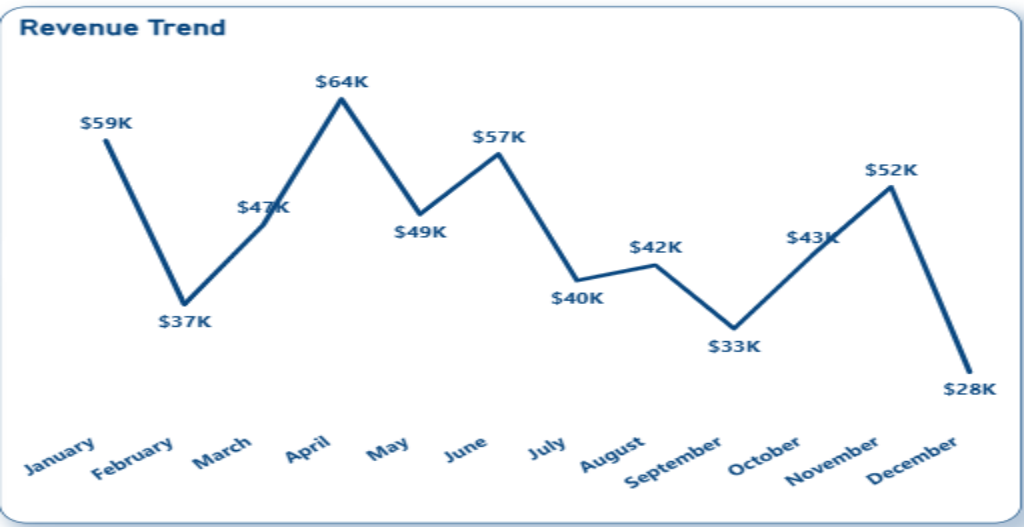
The analysis highlights an important trend: although demand for healthcare services remains consistent, a significant portion of potential revenue is lost due to appointment cancellations, no-shows, and uncollected payments.

This indicates that future growth may not necessarily require acquiring more patients but rather improving the efficiency of existing operations.



7.2 Revenue Trend Analysis

Analysis of monthly revenue trends indicates fluctuations throughout the year. Revenue peaked during April and June before gradually declining in the latter part of the year, reaching its lowest point in December.



Insight

The fluctuations suggest that patient demand and treatment utilization vary considerably throughout the year.

Periods of high revenue are likely driven by increased treatment volume rather than a broad increase in patient registrations.

The decline observed toward year-end may indicate seasonal effects, reduced appointment attendance, delayed treatments, or fewer patient visits.

Business Impact

Revenue inconsistency can create challenges in:

1. Budget planning
2. Workforce allocation
3. Equipment utilization
4. Financial forecasting

Unpredictable revenue patterns may also increase operational costs due to inefficient resource planning.

Recommendation

CareBridge Health Network should adopt a proactive patient engagement strategy designed to maintain service utilization during low-performing months.

Potential initiatives include:

1. Annual health screening campaigns
2. Preventive healthcare packages
3. Follow-up consultation programs
4. Loyalty programs for returning patients
5. Community health awareness initiatives

Implementation Approach

Within the next quarter:

1. Identify months with historically lower patient activity.
2. Launch targeted campaigns 4–6 weeks before these periods.
3. Offer discounted wellness packages to encourage preventive care visits.

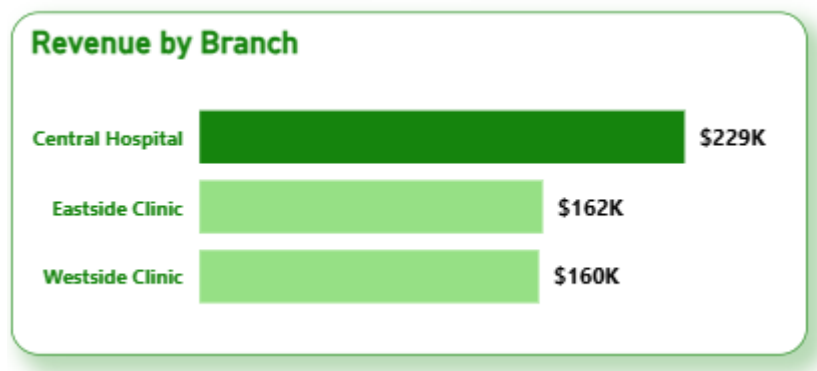
Expected Outcome

Maintaining patient volume during low-performing months could improve revenue stability and increase annual revenue performance.

7.3 Branch Performance Analysis

Branch performance analysis was conducted to evaluate the contribution of each healthcare facility to the overall financial performance of CareBridge Health Network during the period **January–December 2023**.

The analysis indicates that **Central Hospital** generated the highest revenue at approximately **\$229,000**, accounting for about **41.6%** of the network's total annual revenue. **Eastside Clinic** generated approximately **\$162,000**, while **Westside Clinic** contributed approximately **\$160,000**.



Insight

The findings suggest that Central Hospital is the network's strongest-performing facility, indicating higher patient demand, better service utilization, or greater access to high-value medical services. Although Eastside and Westside Clinics generated similar revenue, both lag behind Central Hospital, highlighting opportunities to improve operational performance and service delivery.

Business Impact

Uneven branch performance limits the organization's ability to maximize revenue across its entire network. Underperforming branches may experience lower patient volumes, underutilized resources, and reduced profitability. Improving the performance of these facilities would increase total network revenue while making better use of existing infrastructure.

Recommendation

Management should conduct detailed operational assessments at Eastside and Westside Clinics to identify factors affecting performance. Best practices implemented at Central Hospital should be replicated where applicable, including appointment management, specialist allocation, patient engagement strategies, and marketing initiatives.

Implementation Approach

Short-Term (1–3 Months)

1. Compare patient volumes and appointment completion rates across all branches.
2. Identify operational gaps affecting Eastside and Westside Clinics.

Medium-Term (3–6 Months)

1. Implement successful operational practices from Central Hospital.
2. Increase specialist availability and community outreach where necessary.

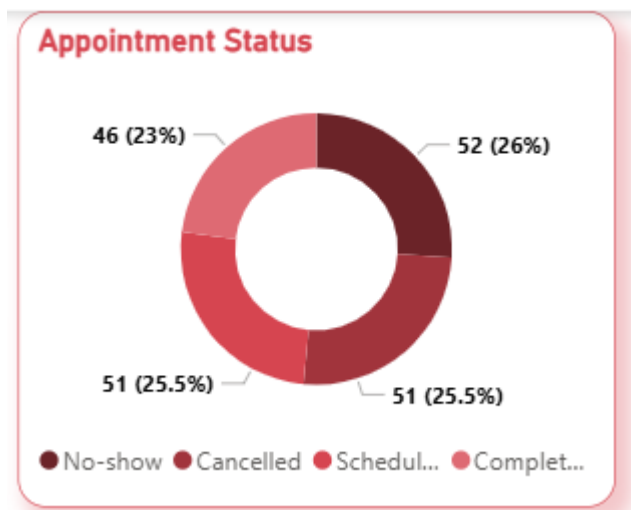
Expected Outcome

Improved branch performance, higher patient volumes, increased revenue generation, and a more balanced contribution from all three healthcare facilities.

7.4 Appointment Status Analysis

Appointment analysis revealed one of the most significant findings within the project.

The appointment completion rate stands at only **23%**, meaning that approximately three-quarters of appointments fail to progress to successful completion.



Insight

This represents a substantial operational inefficiency.

A large number of appointments are categorized as:

1. Scheduled
2. Cancelled
3. No-show

Each missed appointment represents lost revenue and underutilized healthcare resources.

Business Impact

Low appointment completion rates affect:

1. Revenue generation
2. Doctor productivity
3. Patient throughput
4. Service accessibility

Doctors remain underutilized while patients who require appointments may experience scheduling delays.

This issue directly impacts both patient satisfaction and financial performance.

Recommendation

Improving appointment completion rates should be the organization's highest operational priority.

Recommended actions include:

1. Automated SMS reminders
2. Appointment confirmation systems
3. Online rescheduling options
4. No-show monitoring
5. Follow-up communication with absent patients

Implementation Approach

Immediate (0–30 Days)

1. Deploy automated appointment reminders.

Short-Term (1–3 Months)

1. Introduce online appointment confirmations.
2. Implement patient self-service rescheduling.

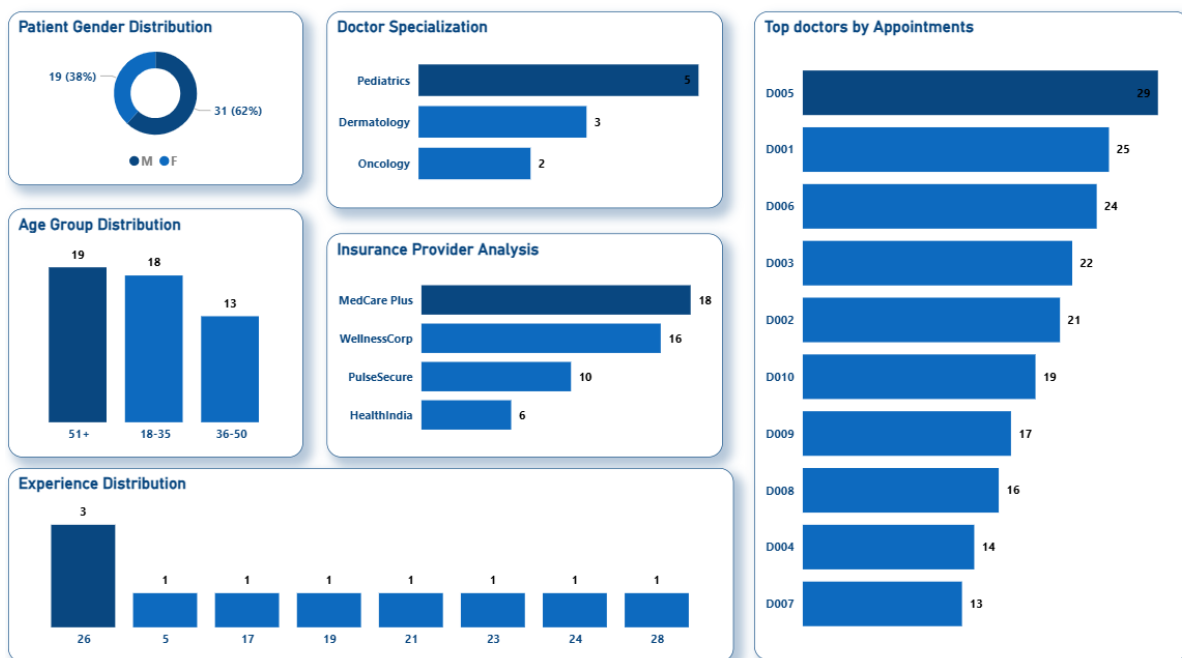
Medium-Term (3–6 Months)

1. Analyze historical attendance patterns.
2. Introduce predictive scheduling techniques.

Expected Outcome

Even a modest improvement in appointment attendance could significantly increase service utilization and revenue without requiring additional staff or infrastructure.

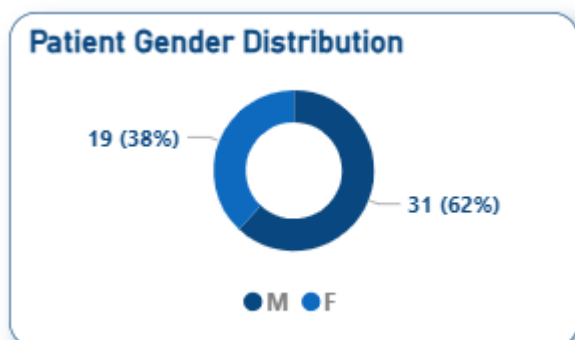
8.0 Patient and Doctor Analytics



8.1 Patient Demographic Analysis

Understanding patient demographics enables CareBridge Health Network to align healthcare services with the needs and characteristics of its patient population.

The analysis reveals that **male patients account for approximately 62% of the patient base**, while **female patients account for 38%**.



Insight

The patient population is predominantly male, indicating that healthcare utilization within the network is higher among men during the reporting period.

This trend may reflect the prevalence of certain health conditions, occupational health concerns, lifestyle-related illnesses, or healthcare-seeking behaviors among male patients within the communities served by CareBridge Health Network.

The imbalance also suggests that female patient engagement may be comparatively lower, presenting an opportunity for the organization to expand outreach efforts and attract a more balanced patient population.

Business Impact

Understanding the dominant patient demographic is critical for service planning, resource allocation, and healthcare program development.

A predominantly male patient population may increase demand for services commonly associated with men's health needs, while simultaneously highlighting underserved opportunities within women's healthcare services.

Failure to address demographic imbalances may limit patient growth opportunities and reduce the network's ability to fully serve the healthcare needs of the broader community.

Recommendation

CareBridge Health Network should continue strengthening services that address the healthcare needs of its primary patient demographic while simultaneously implementing initiatives designed to increase engagement among female patients.

Recommended actions include:

1. Expanding preventive health screening programs targeted at men.
2. Increasing awareness campaigns around cardiovascular health, diabetes, and other conditions commonly affecting male patients.
3. Developing targeted outreach initiatives focused on women's health and wellness.
4. Partnering with community organizations to improve healthcare accessibility and awareness among underrepresented patient groups.

Implementation Approach

Short-Term (1–3 Months)

1. Analyze the most common conditions affecting male patients.
2. Develop targeted health awareness campaigns based on identified trends.

Medium-Term (3–6 Months)

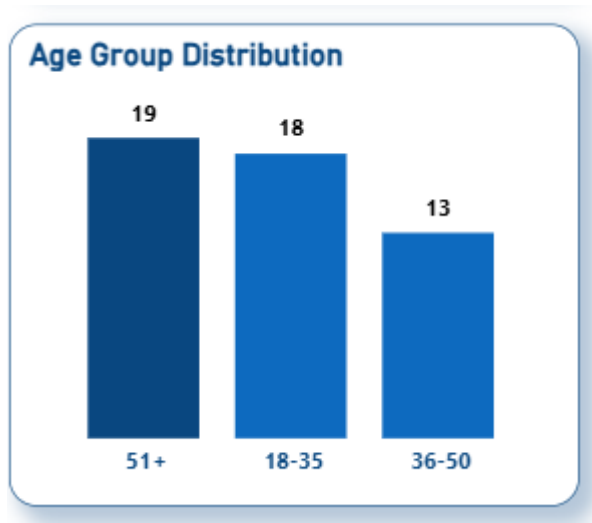
1. Launch women's wellness initiatives and preventive screening programs.
2. Strengthen community engagement efforts to improve healthcare accessibility.

Expected Outcome

By addressing the healthcare needs of both existing and underrepresented patient groups, CareBridge Health Network can improve patient acquisition, strengthen community impact, and create a more balanced and sustainable patient population over time.

8.2 Age Group Analysis

Patients aged 51 years and above represent the largest age segment within the network.



Insight

This indicates that CareBridge Health Network primarily serves an aging population.

Older patients generally require more frequent healthcare interactions, ongoing treatment plans, and long-term disease management.

Business Impact

Demand for healthcare services among older adults is expected to remain high and may continue increasing.

This demographic trend creates opportunities for sustainable long-term growth.

Recommendation

Expand services focused on:

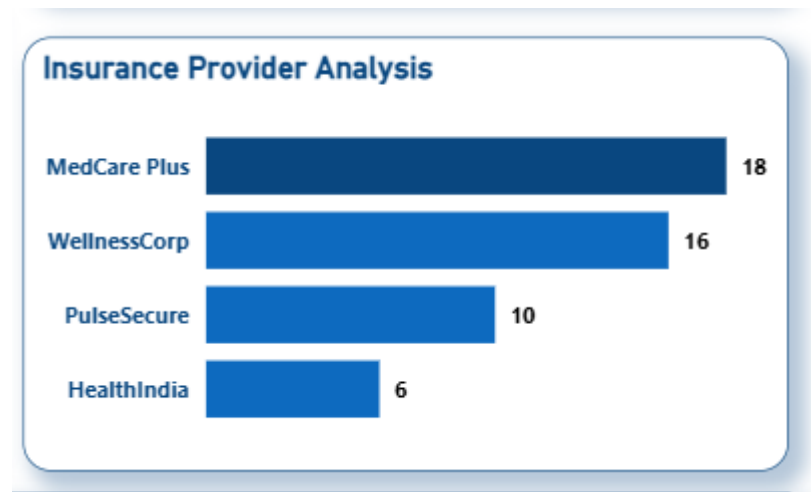
1. Cardiovascular health
2. Diabetes management
3. Chronic disease monitoring
4. Geriatric healthcare programs

Expected Outcome

Increased patient retention and higher recurring revenue from long-term care services.

8.3 Insurance Provider Analysis

The majority of insured patients are concentrated among a limited number of insurance providers, with MedCare Plus and WellnessCorp representing the largest patient shares.



Insight

The organization demonstrates significant dependence on a small number of insurance partners.

Business Impact

Reliance on a limited number of providers creates financial and operational risk.

Changes in reimbursement policies or contract agreements could significantly affect patient volume and revenue.

Recommendation

Diversify insurance partnerships and pursue additional corporate healthcare agreements.

Expected Outcome

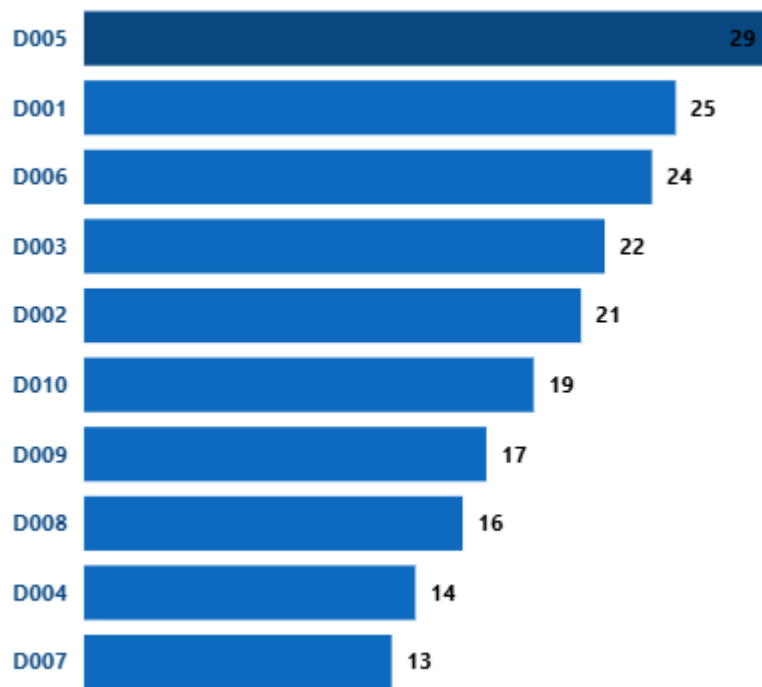
Reduced dependency risk and improved patient acquisition opportunities.

8.4 Doctor Performance Analysis

Analysis of appointment distribution shows that a small number of doctors manage a disproportionately large share of patient appointments.

Doctors D005, D001, and D006 recorded the highest appointment volumes.

Top doctors by Appointments



Insight

Workload distribution appears uneven across the workforce.

While some doctors experience high utilization, others appear relatively underutilized.

Business Impact

Uneven workloads can contribute to:

1. Physician burnout
2. Longer patient waiting times
3. Reduced service quality

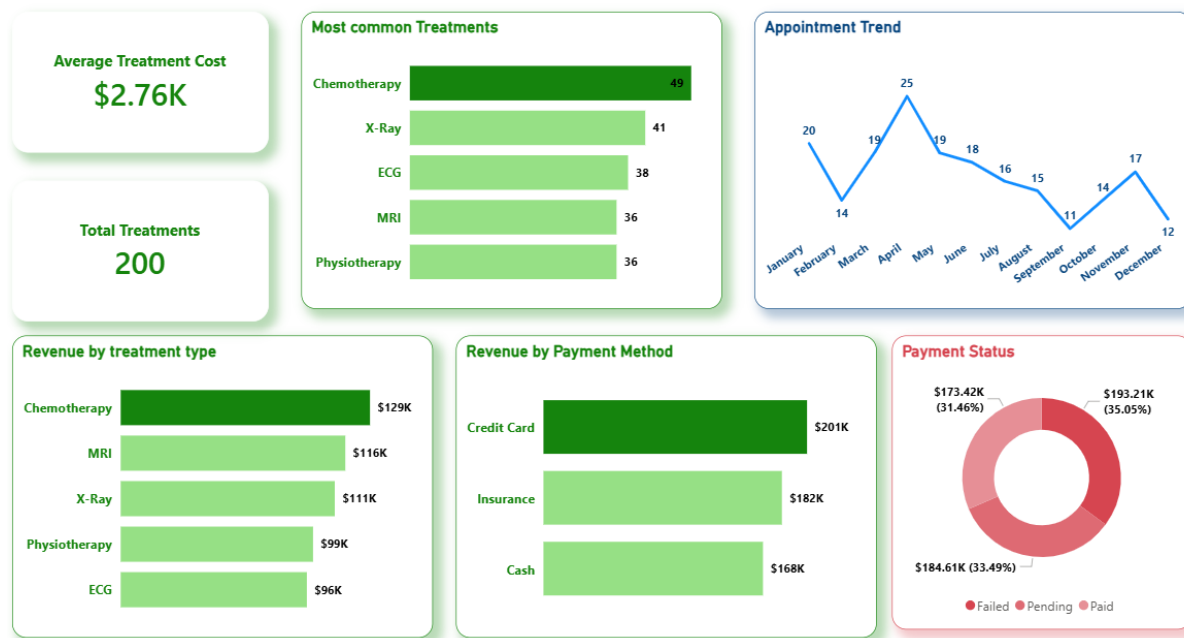
Recommendation

Implement workload balancing initiatives based on specialization demand and appointment volumes.

Expected Outcome

Improved staff productivity, reduced burnout risk, and enhanced patient experience.

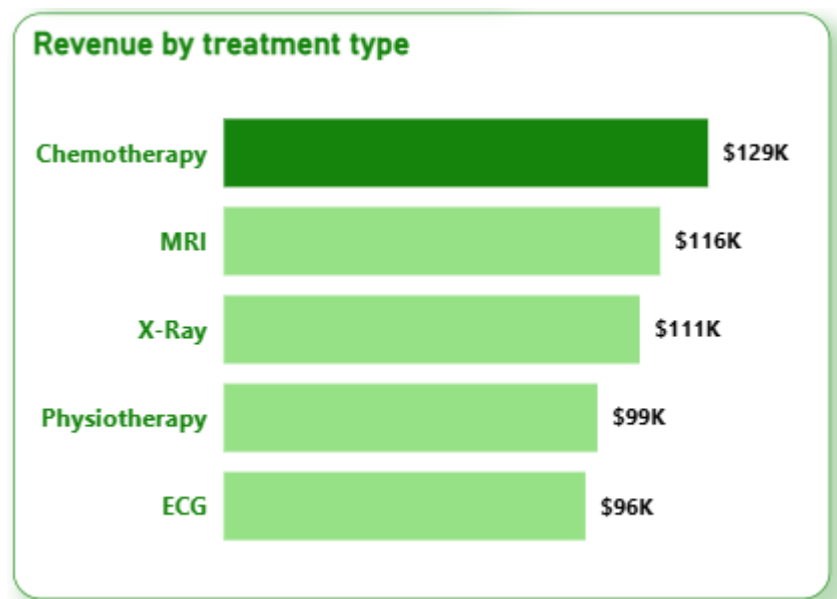
9.0 Financial and Operational Analytics



9.1 Revenue by Treatment Type

Revenue analysis was conducted to identify the medical services contributing most significantly to the financial performance of CareBridge Health Network during **January–December 2023**.

The analysis shows that **Chemotherapy** generated the highest revenue at approximately **\$129,000**, followed by **MRI (\$116,000)**, **X-Ray (\$111,000)**, **Physiotherapy (\$99,000)**, and **ECG (\$96,000)**.



Insight

Revenue generation is concentrated within a few specialized treatment categories, particularly Chemotherapy and advanced diagnostic services. These treatments contribute a substantial

proportion of the hospital's total revenue and therefore represent the organization's most valuable service lines.

Business Impact

High dependence on a limited number of treatment categories increases operational risk. Equipment failure, specialist shortages, or scheduling delays affecting these services could significantly reduce revenue and negatively impact patient care.

Recommendation

CareBridge Health Network should prioritize investment in Chemotherapy, MRI, and X-Ray services by ensuring adequate specialist staffing, preventive equipment maintenance, and efficient appointment scheduling. These services should receive priority when allocating financial and operational resources.

Implementation Approach

Immediate (0–30 Days)

1. Review equipment maintenance schedules for high-revenue services.
2. Assess staffing levels supporting Chemotherapy, MRI, and X-Ray.

Short-Term (1–3 Months)

1. Improve appointment scheduling for high-demand treatments.
2. Minimize equipment downtime through preventive maintenance.

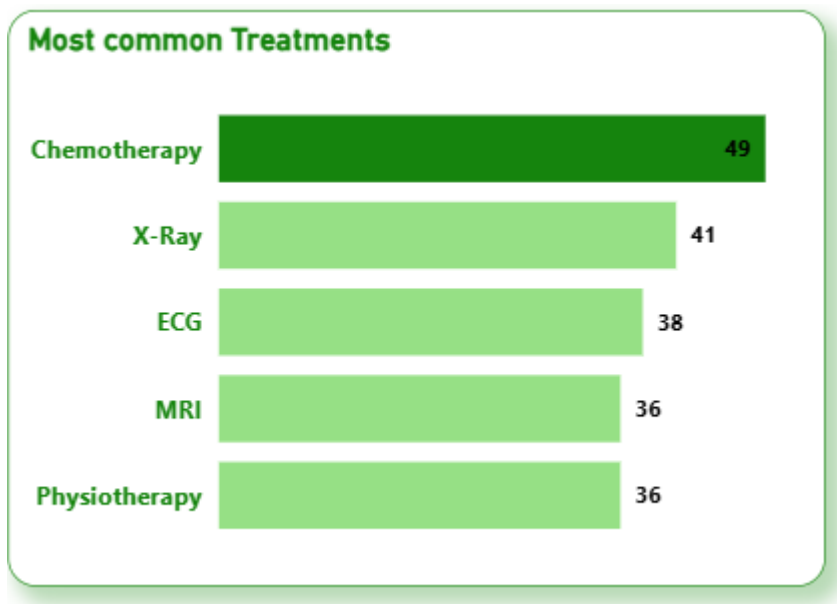
Expected Outcome

Improved service availability, increased treatment capacity, higher patient satisfaction, and sustained revenue growth from the hospital's highest-performing medical services.

9.2 Treatment Utilization Analysis

Treatment utilization analysis was conducted to evaluate the frequency with which different medical services were delivered across CareBridge Health Network during the period **January–December 2023**.

The analysis indicates that **Chemotherapy** was the most frequently performed treatment, followed by **X-Ray**, **ECG**, **MRI**, and **Physiotherapy**. These services accounted for a significant proportion of all treatments delivered during the reporting period, reflecting consistent patient demand for both specialized and diagnostic healthcare services.



Insight

The findings reveal that the most frequently utilized treatments are also among the highest revenue-generating services within the network. This demonstrates a strong relationship between service demand and financial performance, indicating that patients rely heavily on these treatments for ongoing diagnosis, disease management, and specialized care.

The high utilization of Chemotherapy further suggests a sustained demand for oncology services, while the frequent use of diagnostic procedures such as X-Ray, ECG, and MRI highlights their essential role in patient assessment and treatment planning.

Business Impact

High treatment utilization places increased demand on medical equipment, healthcare professionals, and operational resources. Without adequate capacity planning, these high-demand services may experience equipment downtime, longer patient waiting times, appointment backlogs, and increased pressure on healthcare providers.

If not properly managed, these operational challenges could reduce service quality, negatively affect patient satisfaction, and limit the hospital's ability to maximize revenue from its most frequently utilized services.

Recommendation

CareBridge Health Network should continuously monitor treatment demand and align operational resources with utilization trends. Priority should be given to ensuring adequate staffing levels, maintaining diagnostic equipment, optimizing appointment scheduling, and strengthening inventory management for high-demand treatment areas.

The organization should also adopt data-driven capacity planning to anticipate periods of increased demand and allocate resources proactively rather than reactively.

Implementation Approach

Immediate (0–30 Days)

1. Monitor monthly treatment utilization trends across all service categories.
2. Identify services approaching capacity limits.

Short-Term (1–3 Months)

1. Optimize appointment scheduling for high-demand treatments.
2. Strengthen preventive maintenance schedules for frequently used medical equipment.
3. Review staffing requirements for high-utilization service areas.

Medium-Term (3–6 Months)

1. Introduce demand forecasting to support workforce and equipment planning.
2. Expand service capacity where utilization consistently exceeds operational capacity.

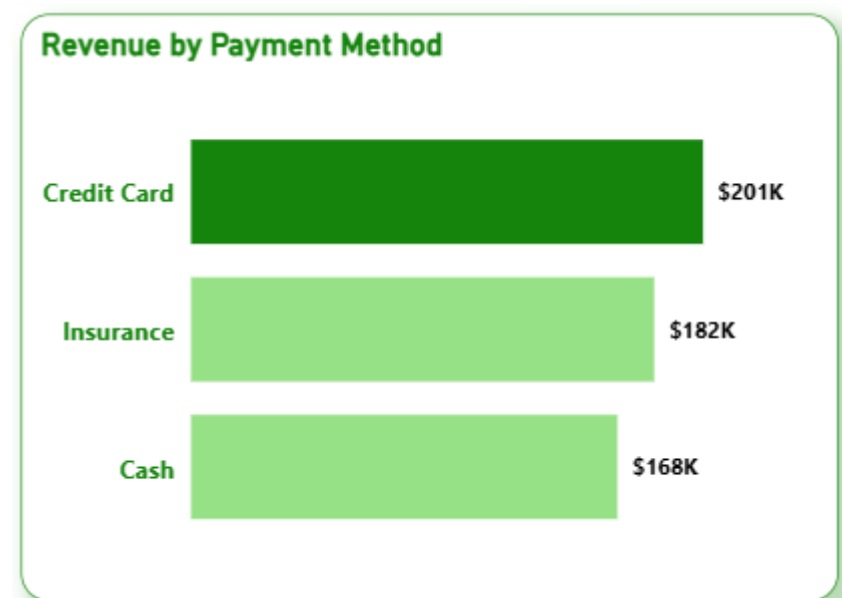
Expected Outcome

Improved operational efficiency, reduced patient waiting times, better utilization of hospital resources, enhanced patient satisfaction, and increased revenue through the consistent delivery of high-demand medical services.

9.3 Payment Method Analysis

Payment method analysis was conducted to understand how patients settled their medical bills and to evaluate the distribution of revenue across available payment channels.

The analysis indicates that **Credit Card** payments generated the highest revenue at approximately **\$201,000**, followed by **Insurance (\$182,000)** and **Cash (\$168,000)**.



Insight

The findings show that electronic and insurance-based payment methods account for the majority of hospital revenue. This indicates increasing patient preference for convenient and structured payment options over traditional cash payments.

Business Impact

A diversified payment structure improves financial flexibility and reduces dependence on a single payment channel. However, continued reliance on insurance reimbursements requires efficient claims processing to avoid payment delays and maintain healthy cash flow.

Recommendation

CareBridge Health Network should continue strengthening digital payment systems while improving insurance claims management. The organization should also encourage wider adoption of electronic payment channels through secure, convenient, and patient-friendly payment solutions.

Implementation Approach

Immediate (0–30 Days)

1. Review the efficiency of current payment processing systems.
2. Monitor insurance claim processing timelines.

Short-Term (1–3 Months)

1. Introduce additional digital payment options.
2. Improve patient awareness of available electronic payment methods.

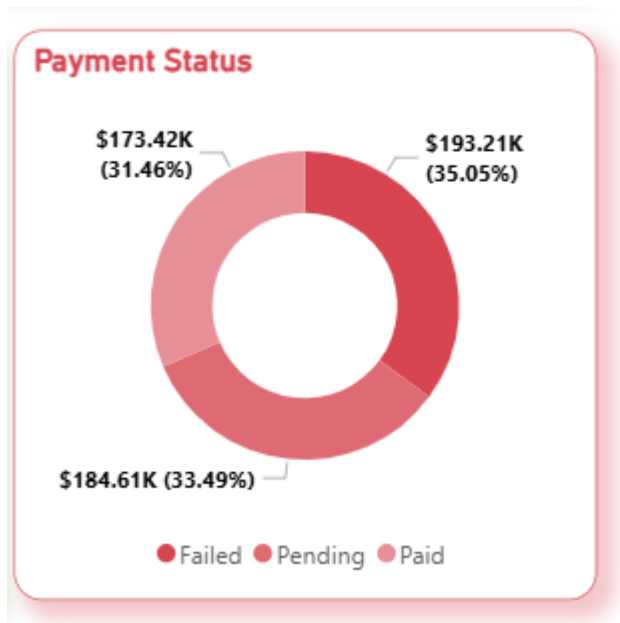
Expected Outcome

Faster payment processing, improved patient convenience, stronger cash flow, and enhanced financial management across the hospital network.

9.4 Payment Status Analysis

One of the most significant findings from the financial analysis relates to payment collection performance.

The dashboard indicates a payment collection rate of approximately **31.46%**, meaning that a substantial portion of generated revenue remains either pending or unpaid.



Insight

While CareBridge Health Network is successful in generating treatment demand and billing activity, it faces challenges converting billed services into collected revenue.

This represents a significant case of revenue leakage.

The organization is effectively providing healthcare services but is not consistently recovering the associated financial value.

Business Impact

Poor payment collection can negatively affect:

1. Cash flow
2. Operational sustainability
3. Equipment investment
4. Staff compensation planning
5. Future expansion initiatives

Even with strong revenue generation, poor collection performance can limit organizational growth.

Recommendation

Improving payment collection efficiency should be considered a strategic priority.

Recommended actions include:

1. Automated billing reminders.
2. Early payment incentives.
3. Follow-up procedures for overdue accounts.

4. Online self-service payment options.
5. Improved insurance claim tracking.

Implementation Approach

Immediate (0–30 Days)

1. Introduce automated payment reminder notifications.

Short-Term (1–3 Months)

1. Establish a structured follow-up process for outstanding balances.

Medium-Term (3–6 Months)

1. Develop a patient self-service billing portal.

Expected Outcome

Improved collection rates, stronger cash flow, and enhanced financial stability.

10.0 Key Findings

The analysis of CareBridge Health Network identified several important trends and business opportunities during the period **January–December 2023**.

Operational Findings

1. The appointment completion rate remains significantly low at **23%**, indicating that a large proportion of scheduled appointments were either cancelled, missed, or left pending.
2. Appointment cancellations and no-shows represent a major source of operational inefficiency and contribute to lost revenue opportunities.
3. Doctor workload is unevenly distributed, with a small number of physicians managing a disproportionately high number of patient appointments.
4. High-demand treatment services such as Chemotherapy, MRI, X-Ray, and ECG require continuous monitoring to ensure adequate staffing, equipment availability, and efficient service delivery.

Patient Findings

1. Male patients constitute the majority of the patient population, accounting for **62%**, while female patients represent **38%**.
2. Patients aged **51 years and above** form the largest age group, indicating a growing demand for chronic disease management and long-term healthcare services.

3. The majority of insured patients are concentrated among a limited number of insurance providers, increasing the organization's dependence on a few reimbursement partners.

Financial Findings

1. CareBridge Health Network generated a total revenue of approximately **\$551.25K** during the reporting period.
2. **Central Hospital** was the highest-performing branch, contributing approximately **\$229K (41.6%)** of the network's total revenue, while Eastside and Westside Clinics generated **\$162K** and **\$160K**, respectively.
3. **Chemotherapy (\$129K), MRI (\$116K), and X-Ray (\$111K)** were the highest revenue-generating treatment categories, highlighting the significant contribution of specialized and diagnostic services to the hospital's financial performance.
4. Revenue generation is concentrated within a small number of high-value treatment categories, making their operational efficiency critical to sustaining financial performance.
5. Payment collection performance remains weak, with a collection rate of only **31.46%**, limiting cash flow despite strong revenue generation.

11.0 Strategic Recommendations

Priority	Recommendation	Timeline	Expected Impact
High	Improve appointment attendance through automated reminders and patient follow-up systems	1–3 Months	Higher appointment completion and increased revenue realization
High	Strengthen payment collection processes through automated billing reminders, online payment platforms, and structured follow-up procedures	1–3 Months	Improved cash flow and financial stability
High	Prioritize investment in Chemotherapy, MRI, and X-Ray services through equipment maintenance, specialist staffing, and scheduling optimization	3–6 Months	Increased treatment capacity and higher revenue generation

Medium	Improve operational performance at Eastside and Westside Clinics by adopting successful practices from Central Hospital	3–6 Months	Balanced branch performance and increased network revenue
Medium	Optimize doctor workload distribution based on appointment demand	3–6 Months	Improved productivity and reduced staff burnout
Medium	Expand healthcare services targeted at older adults, including chronic disease management and preventive care programmes	3–6 Months	Increased patient retention and recurring revenue
Medium	Diversify insurance partnerships and strengthen claims management processes	6–12 Months	Reduced financial risk and improved reimbursement efficiency

12.0 Conclusion

CareBridge Health Network demonstrated strong operational capacity and generated **\$551,250** in revenue across its three healthcare facilities during the **January–December 2023** reporting period. The analysis showed that **Central Hospital** was the network's highest-performing branch, while **Chemotherapy, MRI, and X-Ray** emerged as the primary revenue-generating treatment categories, highlighting the importance of specialized and diagnostic healthcare services to the organization's financial performance.

However, the analysis also identified significant operational challenges that continue to limit overall performance. The **23% appointment completion rate** and **31.46% payment collection rate** indicate substantial revenue leakage, reducing the organization's ability to convert patient demand into realized income. These findings suggest that the hospital's greatest opportunities for growth lie in improving operational efficiency rather than expanding its infrastructure or workforce.

By strengthening appointment management, improving payment collection processes, investing in high-performing treatment services, enhancing lower-performing branches, and optimizing resource allocation, CareBridge Health Network can improve patient outcomes, increase

operational efficiency, strengthen financial performance, and position itself for sustainable long-term growth.